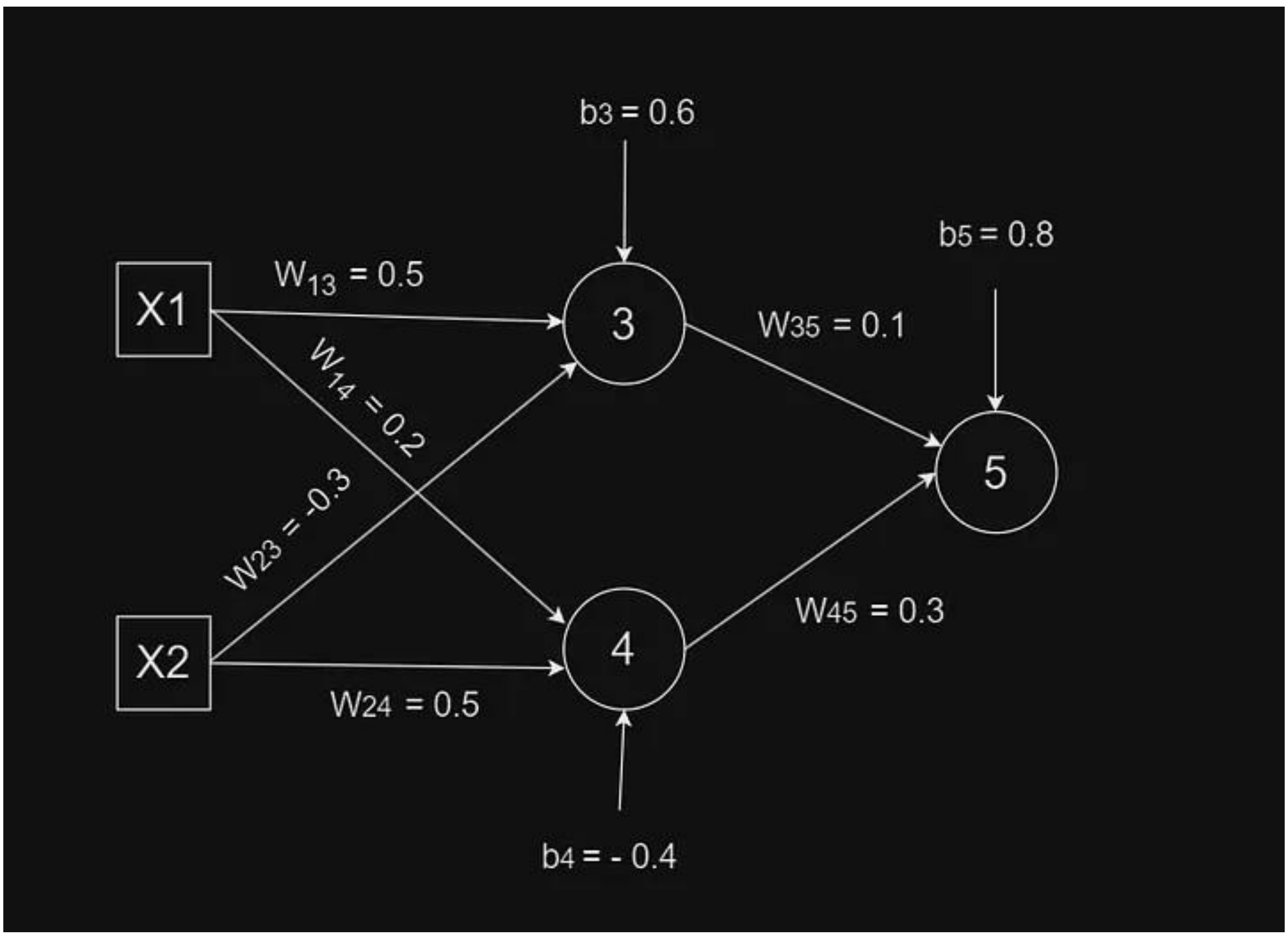
Artificial Neural Network

Q. Consider a *multilayer feed-forward neural network* given below. Let the *learning rate* be *0.5*. Assume initial values of weights and biases as given in the table below. Train the network for the training tuples *(1, 1, 0)* and *(0, 1, 1)*, where last number is target output. Show weight and bias updates by using *back-propagation algorithm*. Assume that *sigmoid activation function* is used in the network.

W13	W14	W23	W24	W35	W45	b3	b4	b5
0.5	0.2	-0.3	0.5	0.1	0.3	0.6	-0.4	0.8

Weight and Bias values



Neural Network

Solution:

Iteration 1

Input and Output values:

X1	X2	Output(τ)
1	1	0

Input and output for Iteration

a. Learning rate (l) = 0.5

b. Activation function = Sigmoid function

$$\left(\text{i.e. } \sigma = \frac{1}{1+e^{-x}}\right)$$

Sigmoid Function

Weight(W) and bias(b) values:

W13	W14	W23	W24	W35	W45	b3	b4	b5
0.5	0.2	-0.3	0.5	0.1	0.3	0.6	-0.4	0.8

Weight and bias values for iteration 1

a. Calculate the input(I) and output(O) for each layer

-For input layer

$O_1 = I_1 = 1$

$O_2 = I_2 = 1$

Input and outputs of Input layer

-For hidden layer

$$I_3 = O_1 \times W_{13} + O_2 \times W_{23} + b_3$$
$$= 1 \times 0.5 + 1 \times (-0.3) + 0.6$$
$$= 0.5 - 0.3 + 0.6$$
$$= 0.8$$

$$O_3 = \frac{1}{1+e^{-I_3}} = \frac{1}{1+e^{-0.8}} = 0.69$$

$$I_4 = O_1 \times W_{14} + O_2 \times W_{24} + b_4$$
$$= 1 \times 0.2 + 1 \times 0.5 + (-0.4)$$
$$= 0.2 + 0.5 - 0.4$$
$$= 0.3$$

$$O_4 = \frac{1}{1+e^{-I_4}} = \frac{1}{1+e^{-0.3}} = 0.5744$$

Inputs and outputs for hidden layer

-For output layer

$$I_5 = O_3 \times W_{35} + O_4 \times W_{45} + b_5$$
$$= 0.69 \times 0.1 + 0.5744 \times 0.3 + 0.8$$
$$= 0.0690 + 0.1723 + 0.8$$
$$= 1.0413$$

$$O_5 = \frac{1}{1+e^{-I_5}} = \frac{1}{1+e^{-1.0413}} = 0.7391$$

Input and output of output layer

b. Calculation of error at each node:

$$\boxed{Err_j = O_j(1 - O_j)(T_j - O_j)}$$

Error calculation formula for output layer

- ***For output layer:***

  $Err_5 = O_5\,(1 - O_5)\,(\tau - O_5)$

  $= 0.7391\,(1 - 0.7391)\,(0 - 0.7391)$

  $= 0.7391\,(0.2609)\,(- 0.7391)$

  $= - 0.1425$

Error calculation for output layer

$$\boxed{Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk}}$$

Error calculation formula for hidden layer

- ***For hidden layer:***

$$
\begin{aligned}
Err_4 &= O_4\,(1 - O_4)\,(Err_5 \times W_{45}) \\
&= 0.5744\,(1 - 0.5744)\,(-\,0.1425 \times 0.3) \\
&= 0.5744\,(0.4256)\,(-\,0.04275) \\
&= -\,0.010451
\end{aligned}
$$

$$
\begin{aligned}
Err_3 &= O_3\,(1 - O_3)\,(Err_5 \times W_{35}) \\
&= 0.69\,(1 - 0.69)\,(-\,0.1425 \times 0.1) \\
&= 0.69\,(0.31)\,(-\,0.1425) \\
&= -\,0.030481
\end{aligned}
$$

c. Update the weight

$$\boxed{W_{ij}(new) = W_{ij}(old) + l \times O_i \times Err_j}$$

Formula for new weight calculation

$$W_{45}(new) = W_{45}(old) + l \times O_4 \times Err_5 = 0.3 + 0.5 \times 0.5744 \times (-\,0.1425) = 0.2591$$
$$W_{35}(new) = W_{35}(old) + l \times O_3 \times Err_5 = 0.1 + 0.5 \times 0.69 \times (-\,0.1425) = 0.0508$$

$$W_{24}(new) = W_{24}(old) + l \times O_2 \times Err_4 = 0.5 + 0.5 \times 1 \times (-\,0.010451) = 0.494775$$
$$W_{14}(new) = W_{14}(old) + l \times O_1 \times Err_4 = 0.2 + 0.5 \times 1 \times (-\,0.010451) = 0.194774$$

$$W_{23}(new) = W_{23}(old) + l \times O_2 \times Err_3 = -\,0.3 + 0.5 \times 1 \times (-\,0.030481) = -\,0.315240$$
$$W_{13}(new) = W_{13}(old) + l \times O_1 \times Err_3 = 0.5 + 0.5 \times 1 \times (-\,0.030481) = 0.484759$$

Updated weight values

d. Update the bias value

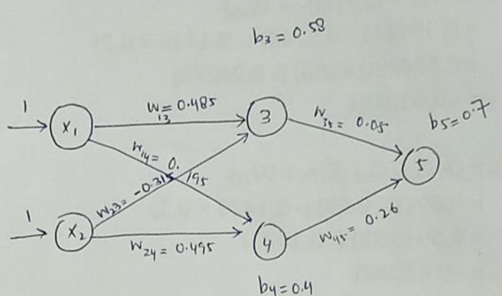
$$b_j = b_{j(old)} + l \times Err_j$$

Formula for new bias calculation

$$b_3 = b_{3(old)} + l \times Err_3 = 0.6 + 0.5 \times (-\ 0.030481) = 0.584759$$
$$b_4 = b_{4(old)} + l \times Err_4 = -\ 0.4 + 0.5 \times (-\ 0.010451) = -\ 0.405225$$
$$b_5 = b_{5(old)} + l \times Err_5 = 0.8 + 0.5 \times (-\ 0.1425) = 0.7288$$

Updated bias values

e. *Again feed forward by updated weights and calculate error at O_5. O_5 = 0.314 has less value comparatively than previous iteration that was 0.7391. our desired value is 0. Err_5 previously was -0.1425 after updating weights and biases it is now -0.067.*



Updated waits

$$l_3 = O_1 \times w_{13} + O_2 \times w_{23} + b_3$$
$$= 1 \times 0.485 + 1 \times (-0.315) + 0.58$$
$$= 0.485 - 0.315 + 0.58 = 0.75$$
$$O_3 \Rightarrow 0.32$$

$$l_4 = O_1 \times w_{14} + O_2 \times w_{24} + b_4$$
$$= 1 \times 0.195 + 1 \times 0.495 + 0.4 = 1.09$$
$$O_4 = 0.25$$

$$l_5 = O_3 \times w_{35} + O_4 \times w_{45} + b_5$$
$$= 0.32 * 0.05 + 0.25 * 0.26 + 0.7$$
$$= 0.016 + 0.065 + 0.7$$
$$\Rightarrow 0.781$$
$$O_5 = 0.314$$

$$\text{Error at } O_5 = O_5 (1 - O_5)(T - O_5)$$
$$= 0.314 (1 - 0.314)(0 - 0.314)$$
$$= 0.314 (0.686)(-0.314)$$
$$= -0.067$$